Investigating Data Representations in "Year-End Wrap Ups"

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Abstract

The field of visualization research has yet to explore how company year-end wraps present their data to users. This work investigates this phenomena to find which chart types are used most often to effectively communicate year-end reflections. We found that *Tables*, which also include ranked lists and numeric values, are the most used by companies in year-end wraps, with all 26 providers using them in some capacity. We found that Cloudflare had the most chart types used out of any provider with 8 chart types used. We then compared this to the results from similar investigations, finding some correspondence. However, likely due to the role of year-end wraps, tables more noticeably dominated other chart types.

1 Introduction

Year-end wraps are a way of companies presenting data to users though infographics and other mediums to share statistics and communicate year-end reflections. In recent years, more companies are taking part in year-end recaps making them a more common form of user centered data communication. This study aims to look at year-end wraps from 2025 to analyze the chart types used by companies to display that information. The dataset includes providers like Spotify, YouTube, Letterboxd, Cloudflare and more. We reviewed existing research to see if there was any studies done before that looked at year-end recaps, but found no exact results. The closest items were from Rohwer et al. [6] where they talk about weather data representations on a watch OS. Because no existing research existed, we looked at each year-end recap from the providers and aimed to see what chart types appears the most often. The chart types we used were area, bar, circle and more.

2 Related Work

Islam et al. [3] described the use of visualization in watches and the types of data and representations that are displayed on them. They found 237 smartwatch users and looked at the predominant display on the watches. Participants had health and fitness data displayed on the watch faces the most. Kim et al. [4] discussed the determining factors in what makes a visualization inviting or interpretable in articles. Their study revealed that thumbnails and thumbnail design components can be used to leverage the attraction of readers, allowing them to understand main points and context in articles better. Thus, through the inclusion of thumbnails there is an untapped space to generate or recommend ideal visual components.

Rahman et al. [5] examined the importance of annotations in information visualizations, showing their importance in improving the understanding of visual data. They looked at the types, design spaces, and generation methods across tolls using annotations. Their results looked at improving tool development through their finding to help with future research. Rohwer et al. [6] investigated the use of visualization in android *Wear OS* smartwatches and the type of data that is displayed on them through different weather apps. They looked at 15 different data types and 11 weather applications. They found that the temperature and weather condition were the most viewed and the results were similar to Islam et al. [3]. Other researchers have looked at the taxonomy of chart types, such as Borkin et al. [1], , which we will later use to condense our findings into more standardized categories.

3 Study Design

The data collected was from real sources found via online searches. We listed terms to find year-end wraps and links to each wrap we found in an excel document to help create initial tags. The tags were as follows: *"year-end wrap"*, *"yearly summary"*, *"year-in review"*, *"wrapped"*, *"end of year recap"*, *"best of 2025"*, *"year in highlights"*, *"year in trends"* and *"end of year report"*. Once we found as many year-end recaps as we could, each of us individually searched the pages for charts, with those found shown in Table 1.

During this step we used a top-down approach to qualitative coding. Once we had our list of if each provider used the chart type, an *X* was placed if used and left blank if not used. We discussed our findings until we had full agreement. Some of our disagreements were on if column chart and bar chart should be separated or combined into one chart type. We agreed on the categories shown in Table 2 as the charts used from providers in year-end wraps. We then did another session of coding with these agreed-upon chart types. After reaching full agreement with these results, we condensed our chart types into categories defined from Borkin et al. [1] to have a more concise and standardized approach. In summary, we started with 19 chart types originally, went down to 17 after our first discussion and then finished with 12 for our final table.

4 Data Analysis and Results

After the collection of the data we had a good base set of what information was present in year-end wraps and what chart types were used the most. Table 3 was created using the categories from Borkin et al. [1], which shows how subcategories were changed to single chart types. This is how we were able to take Table 4 and condense it into Table 5. This lead to us being able to find chart type frequencies between all providers. Listed on Table 5 is the total number of charts used from each provider and the total number of charts used in total. We were able to find which chart types were the most used and least used from this. The most used chart type was table with all providers using it. Most of the company year-end wraps focused on infographic visuals or ranked lists to present their data leading to the large disparity between table and every other chart type. The next closest chart type was bar chart with only 6 providers using them or 23%, compared to table with 100%. The least used chart types were text and grid and matrix with zero uses from all providers.

Most year end wraps used ranked lists to provide users with year-end data with Cloudflare being one of the few providers to use many chart types. The provider with the most amount of charts used was Cloudflare with 8/12 total charts used. The next providers with the most charts were Wikipedia, eBird and Stack Overflow with 4/12 charts used each. There was a large tie with other providers having only one chart used each. Those being Chess, Tinder, Strava, Collector Daily, Dead by Daylight, Pocket Casts, Spotify, Google, Statsig, YouTube, TikTok, Netflix, UDisc, NYT Games, Eight Sleep,

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Table 1: Initial Set of Categories

Line Chart	Multi-series Line Chart	Choropleth Map	Tree Map	Heat Map
Column Chart	Bar Chart	Grouped Bar Chart	Pictorial Bar Chart	Hilbert Curve
Stacked Area Chart	Pie Chart	Table Chart	Numeric	Ranked List
Ranked List Visual	Categorical Graphic	Multi-panel Dashboard	Comparison Panel	

Table 2: Final Set of Categories

Bar Chart	Bubble Map	Choropleth Map	Column Chart	Donut Chart
Dot Plot	Dumbbell Chart	Grouped Bar Chart	Icon Array	Line Chart
Map	Multi-series Line Chart	Numeric	Pie Chart	Radial Chart
Ranked List	Sankey Diagram	Stacked Area Chart	Stacked Bar Chart	Stacked Prop. Bar Chart
Table	Timeline	Tree Map		

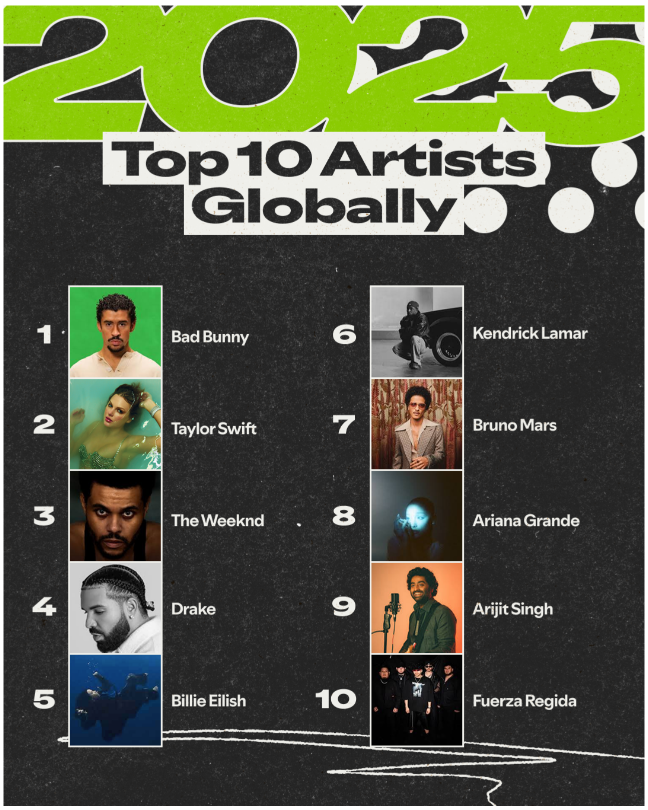

Figure 1: Spotify Wrap Ranked List

Oura and Letterboxd. These companies provided single chart types in their year-end wrap ranking items like top artists, most played songs, top movies of the year and so.

We then compared these results to Deng et al. [2], shown in Table 6. Compared to their findings when looking at 490 visualizations, they found that 33% of infographics used table and 31% used diagram. Other chart types filled in the rest of the percentage with much smaller margins with map being the next highest with 9%. These results partly align with our results. Table was the most used graphic with 52% of all providers using them for year-end wraps, but diagram was not used nearly as much with only 4% of providers using it. Our next highest reporting of chart types was bar chart with 12%, or 6 providers using it. Compared to MassViz with 6% and VisImages 13%. This was only a portion of visualization under the mass visualization section with scientific, news, government and infographics each having a section totaling 10,289 visualization reports. When comparing all sections together under the VisImages section, tables percentage drops to 11% of total charts used and diagram to 7%. These percentages very depending on which one is looked at. Diagram is more in line with the average generated through all media types with their 7% compared to our 4%. Table is no longer close to our findings with 11% compared to our 52%.

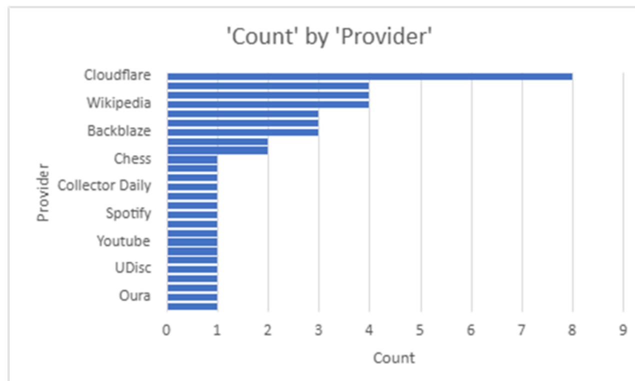

Figure 2: Total Count of Chart Types

5 Discussion

The information we found in our data analysis was in line with our thought process going into this research. Most year-end wraps are ranking systems to find what are the top results for that year. This is in line with our findings as table was the most used chart type with all providers using it. Table included ranked list, numeric and table chart types. These were the most common charts found in all year-end wraps. These kind of infographics using ranked list make the most sense. Companies try to keep year-end wraps simple so that anyone can follow along and enjoy seeing what was the most popular item that year. This simplicity is what is the most crucial factor in creating year end wraps. Reflective visualizations have a lot of summative statistics, comparisons and straightforward presentations leading to these more basic chart types. But that doesn't mean that all year-end wraps adhere to that notion. Cloudflare, Stack Overflow, eBird and Wikipedia all had more summative statistics than just ranked list, providing more information to users.

These observations lead to the idea that these more advanced year end wraps are directly correlated to the audience trying to be reached. The more advanced the website, like Cloudflare, the more information present in more forms. Most providers are not trying to inform deeply in their wraps, they are providing this information to reinforce brand types, showcase growth or highlight success that they've had throughout the year leading to the more simplistic chart types. The limitations of these simplistic chart types is that the more advanced data that other users would like to see is

Table 3: Chart Categories with Subtypes

Chart Type	**Included Charts**					
Area	Stacked Area Chart					
Bar	Bar Chart (H)	Column Chart (V)	Grouped Bar Chart	Stacked Bar Chart	Stacked Prop. Bar Chart	Radial Chart
Circle	Pie Chart	Donut Chart				
Diagram	Timeline	Sankey Diagram				
Distribution	Icon Array					
Grid & Matrix						
Line	Line Chart	Multi-Series Line Chart				
Map	Map	Choropleth Map				
Point	Dot Plot	Bubble Map	Dumbbell Chart			
Table	Table	Ranked List	Numeric			
Text						
Trees & Networks	Tree Map					

Table 4: Individual Chart Types Dataset

Name of Provider	Bar Chart (H)	Bubble Map	Column Chart (V)	Choropleth Map	Grouped Bar Chart	Icon Array	Line Chart	Multi-Series Line Chart	Numeric	Pie Chart	Ranked List	Stacked Area Chart	Table	Timeline	Tree Map	Stacked Bar Chart	Stack Prop. Bar Chart	Map	Donut Chart	Radial Chart	Dumbell Chart	Sankey Diagram	Dot Plot	Total	Percentage
Backblaze			X				X		X	X			X											5	21.7%
Cloudflare	X	X		X			X	X	X		X	X	X		X	X							X	12	52.2%
Duolingo				X					X		X		X											4	17.4%
Eight Sleep			X						X		X		X											4	17.4%
Garmin			X		X			X	X															4	17.4%
Google											X													1	4.3%
Letterboxd									X		X													2	8.7%
Netflix			X						X															2	8.7%
NYT Games			X						X		X													3	13.0%
Oura	X								X		X		X											4	17.4%
Pocket Casts									X		X		X											3	13.0%
Spotify									X		X													2	8.7%
Strava									X															1	4.3%
Tinder									X															1	4.3%
Tiktok									X		X													2	8.7%
UDisc									X															1	4.3%
Wikipedia						X			X					X										3	13.0%
Youtube									X		X													2	8.7%
Dead By Daylight									X															1	4.3%
eBird				X			X		X		X							X						5	21.7%
Chess									X		X		X											3	13.0%
Games Industry											X		X						X					3	13.0%
Epic Games						X			X		X									X				4	17.4%
Collector Daily									X				X											2	8.7%
Stack Overflow	X								X				X				X				X	X		6	26.1%
Statsig	X				X				X															3	13.0%
Total	4	1	5	3	2	2	3	2	24	1	15	1	10	1	1	1	1	1	1	1	1	1	1		
Percentage	15.4%	3.8%	19.2%	11.5%	7.7%	7.7%	11.5%	7.7%	92.3%	3.8%	57.7%	3.8%	38.5%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%	3.8%		

Table 5: Final Dataset

Name of Provider	**Area**	**Bar**	**Circle**	**Diagram**	**Distribution**	**Grid & Matrix**	**Line**	**Map**	**Point**	**Table**	**Text**	**Trees & Networks**	Total	Percentage
Backblaze			X				X			X			3	25.0%
Cloudflare	X	X	X				X	X	X	X		X	8	66.7%
Duolingo		X						X		X			3	25.0%
Eight Sleep										X			1	8.3%
Garmin							X			X			2	16.7%
Google										X			1	8.3%
Letterboxd										X			1	8.3%
Netflix										X			1	8.3%
NYT Games										X			1	8.3%
Oura										X			1	8.3%
Pocket Casts										X			1	8.3%
Spotify										X			1	8.3%
Strava										X			1	8.3%
Tinder										X			1	8.3%
Tiktok										X			1	8.3%
UDisc										X			1	8.3%
Wikipedia		X		X	X					X			4	33.3%
Youtube										X			1	8.3%
Dead By Daylight										X			1	8.3%
eBird		X					X	X		X			4	33.3%
Chess										X			1	8.3%
Games Industry			X							X			2	16.7%
Epic Games		X			X					X			3	25.0%
Collector Daily										X			1	8.3%
Stack Overflow		X		X					X	X			4	33.3%
Statsig										X			1	8.3%
Total	1	6	3	2	2	0	4	3	2	26	0	1		
Percentage	3.8%	23.1%	11.5%	7.7%	7.7%	0.0%	15.4%	11.5%	7.7%	100.0%	0.0%	3.8%		

Table 6: Dataset Comparison

	Area	**Bar**	**Circle**	**Diagram**	**Distribution**	**Grid & Matrix**	**Line**	**Map**	**Point**	**Table**	**Text**	**Trees & Networks**
VisImages	**4%**	**13%**	2%	7%	**4%**	**8%**	**12%**	7%	**12%**	11%	**1%**	**18%**
MassVis (Infographics)	**4%**	6%	5%	**31%**	0%	2%	2%	**9%**	3%	33%	0%	6%
Our Dataset	2%	12%	**6%**	4%	**4%**	0%	8%	6%	4%	**52%**	0%	2%

not displayed so that more people can understand and contribute to discussion around the wraps. This somewhat aligns with Deng et al. [2], as table was the most used chart type for infographics. For very specifically year-end wraps there would be a large disparity between table and all other chart types as most year-end wraps using ranked list to display information to users.

6 Future Work

As for future work discussing year-end wraps, there are two routes we figured could be explored more. The first is searching more intently for company year-end wraps that we were unable to find and compare them to the existing results we gathered to see if our observed patterns continue. The other would be looking at audience and how they play a key factor in how company year-end wraps are generated. The main idea to be looked at would be whether, based on the audience trying to be reached, more complex data visualizations would be used. We found this through Cloudflare, which had more tech-driven audience and, because of that, provided more complex visualizations than other providers.

7 Conclusion

In this paper, we looked at different company generated year-end wraps and the amount of chart types used by each provider. We found that the most used chart type was table with all providers using it, followed by bar chart with only 6 providers using it. We also found that Cloudflare had the most amount of chart types used from a provider with 8 chart types used out of 12. The next providers with the most chart types were Stack Overflow, eBird and Wikipedia with 4 uses each. When we compare these results to Borkin et al. [1] (MassVis) and Deng et al. [2] (VisImages: Infographics) we found that these results somewhat vary. Table was the most used Infographic with 33% percent of 490 visualizations used which is in line with our findings for year-end wraps as all providers used the table chart type in their reports. Diagram however was not used as much as it was in their report with only 4%, or 2 providers using it compared to their 31%. These results do not share a clear correlation as our research explored year-end wraps which are generally more ranked based and not broad visualizations in general.

As for the total VisImages, out of 10,289 visualizations used, their results were more spread out with trees & networks holding the highest percentage used (18%). Table had 11% of total visual images used, which does not align with our findings as our results show a clear dominating trend through table charts. This makes sense compared to our findings, as all year-end wraps feature ranking in some capacity leading to this large disparity of chart types.

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